AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM129
Technologies in Practice

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 20/Apr/2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2 h

Instructions:

- Total Marks:**60**
- This exam consists of **6** Questions.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Question 1: Choose the correct answer [10 Marks]

1. **What does the OSI model stand for?**
   a) Open Systems Interconnection
   b) Operating Systems Interface
   c) Online Systems Integration
   d) Optical Systems Interconnection
2. **Which of the following is NOT a type of operating system?**
   a) Batch Operating System
   b) Time-Sharing Operating System
   c) Distributed Operating System
   d) Network Routing Operating System
3. **What is the purpose of a default gateway in a network?**
   a) To connect devices within the same LAN
   b) To forward traffic from a local network to other networks
   c) To provide wireless connectivity
   d) To manage DNS queries
4. **Which device primarily operates at the Transport Layer?**
   a) Switch
   b) Router
   c) PC
   d) Bluetooth
5. **What does TCP/IP stand for in networking?**
   a) Transmission Control Protocol/Internet Protocol
   b) Transfer Control Process/Information Protocol
   c) Terminal Communication Protocol/Internet Process
   d) Transport Communication Protocol/Interface Protocol
6. **What is the primary purpose of ARP (Address Resolution Protocol)?**
   a) To map IP addresses to MAC addresses
   b) To resolve domain names to IP addresses
   c) To assign IP addresses dynamically
   d) To encrypt network traffic
7. **Which OSI model layer handles logical addressing?**
   a) Physical Layer
   b) Data Link Layer
   c) Network Layer
   d) Application Layer
8. **What is the purpose of a subnet mask in IP addressing?**
   a) To encrypt data packets
   b) To identify the network portion of an IP address
   c) To assign MAC addresses
   d) To manage DNS queries
9. **Which of the following is a characteristic of a distributed operating system?**
   a) Runs on a single machine only
   b) Shares resources across multiple machines
   c) Does not support multiprocessing
   d) Used only for batch processing

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10. **Which of the following is NOT a feature of a real-time operating system?**
    a) Predictable response times
    b) Support for multitasking
    c) Batch processing of tasks
    d) Used in embedded systems

Question 2: Compare between the following [10 Marks]

a) Compare between IPv4 and IPv6? (2 Marks)
b) Compare between Coaxial cable and Fiber optic cable? (2 Marks)
c) Compare between POP3 and IMAP4? (2 Marks)
d) Compare between Private cloud and Public cloud? (2 Marks)
e) Compare Malware and Spyware? (2 Marks)

Question 3: Answer the following questions [10 Marks]

a) A small business with 10 employees is setting up their first network. They need to ensure that their workstations, printers, and servers can all communicate effectively. They also want to minimize network configuration errors. How should they assign IP addresses, and what equipment would be necessary to ensure smooth communication between devices? (4 Marks)
b) Explain about unicast transmission, broadcast transmission, and multicast transmission each with an example diagram. (6 Marks).

Question 4: Answer the following questions [10 Marks]

a) Match the following components with their roles. (4 Marks)

1. IaaS (Infrastructure as a Service	a) An operating system structure that moves functions from the kernel into user space, with communication between modules handled by message passing.
2. Fog Computing	b) A newer form of social engineering that utilizes Voice over IP (VoIP) technology to deceive victims into releasing confidential information.
3. Vishing	c) Processes data closer to the network edge, reducing the burden on networks and enhancing real-time decision-making.
4. Microkernel	d) A cloud offering where the provider supplies IT managers access to physical network infrastructure, storage, and virtualized network resources.

b) Name and define three distinct types of malicious software (malware) (3 Marks)

c) Explain the purpose of port numbers in network communication and define the three categories of port numbers as assigned by ICANN. (3 Marks)

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Question 5: Answer the following questions [10 Marks]

a) List and explain any four common security threats and vulnerabilities that networks face, and how can they be mitigated? (4 Marks)

b) If a company is assigned the IP address 192.168.45.12. Answer the following questions. (4 Marks)
   1) Identify the class of this IP address.
   2) What is the default subnet mask for this class?
   3) Determine the network portion and host portion of the address.
   4) Calculate the maximum number of hosts supported in this class by default.

c) Explain how bandwidth influence the performance of a network, and what are its primary units of measurement? (2 Marks)

Question 6: Answer the following questions [10 Marks]

a) What is a microkernel, and what are its benefits? (3 Marks)
b) What is the purpose of a linker and a loader in an OS? (3 Marks)
c) Define caching. What are the benefits of caching? (4 Marks)

End of Questions

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